

**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,  
RUMOMASI**

TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: SEVEN

CLASS: SS1

SUBJECT: Agricultural Science

## **TOPIC: Soil Fertility & Fertilizers**

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### **Introduction**

- Soil fertility is the ability of the soil to supply the nutrients needed for plant growth in the right amounts. A fertile soil is rich in humus, well aerated, holds water and has the right pH.

### **Plant nutrients**

- **Macronutrients:** nitrogen (leaf growth), phosphorus (roots and energy), potassium (flowering, water balance), calcium, magnesium and sulphur. **Micronutrients:** iron, zinc, copper, boron, manganese, molybdenum — needed in traces. **Deficiency symptoms:** yellowing (N), purple leaves (P), poor flowering (K).

### **Types of fertilizers**

- **Organic** — farmyard manure, compost, green manure, poultry droppings: improve soil structure and humus, release nutrients slowly.
- **Inorganic (artificial)** — NPK, urea, sulphate of ammonia, single superphosphate, muriate of potash: concentrated, fast-acting, but can acidify the soil and pollute water.
- The appropriate fertilizer depends on the soil test result and the crop.

### **Maintaining fertility**

- Crop rotation (legumes fix nitrogen), application of manure and compost, liming acid soils, contour ploughing and cover cropping to prevent erosion, and fallowing.

### **CLASS EVALUATION**

1. Define soil fertility.
2. Name two macronutrients and their functions.
3. Differentiate between organic and inorganic fertilizers.
4. How do legumes improve soil fertility?

**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,  
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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: EIGHT

CLASS: SS1

SUBJECT: Agricultural Science

## **TOPIC: Land Use & Farm Planning**

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### **Introduction**

- Land use is the way land is utilised (farming, forestry, housing, industry); farm planning is deciding how to use the farm's land, labour, capital and time for maximum profit.

### **Principles of land use**

- Match the use to the physical characteristics (soil, slope, rainfall); keep the best land for arable crops, the steeper slopes for pasture or forest; zone the farm (buildings on dry, accessible land); leave paths and boundaries; control erosion on slopes with contour lines and cover.

### **Farm planning**

- Choose enterprises (crop, livestock, mixed) by market demand, climate, capital and skill; draw a farm map showing the layout of fields, buildings, roads and water; prepare a seasonal calendar of operations (land preparation, planting, weeding, harvesting); and budget costs and expected returns.

### **Benefits**

- Better use of resources, less erosion, easier management, higher yields and profit, and a plan that can be followed and evaluated.

### **CLASS EVALUATION**

1. Define land use.
2. What factors should guide farm planning?
3. Why is zoning important on a farm?
4. Draw a simple sketch of a farm layout with buildings, fields and roads.

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: NINE

CLASS: SS1

SUBJECT: Agricultural Science

## **TOPIC: Crop Production I (Land Preparation & Planting)**

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### **Introduction**

- Crop production is the cultivation of plants for food, feed, fibre and industrial use. The steps follow a sequence: site selection, land preparation, planting, maintenance and harvesting.

### **Land preparation**

- Clearing (manual or mechanical), stumping (removing roots), ploughing (turning the soil), harrowing (breaking clods), and making ridges, beds or mounds (for yam, cassava, maize on ridges; vegetables on beds). Soil is prepared to give roots a loose, well-drained medium and to bury weeds.

### **Planting**

- Method: direct sowing (maize, beans, groundnut) or transplanting (tomato, pepper, vegetables from nursery).
- Spacing: the distance between plants and rows depends on the crop — maize 75 cm × 30 cm, yam 1 m × 1 m; correct spacing prevents overcrowding and competition.
- Depth: seeds sown at about 2–3 times their diameter; time: with the rains, early enough for a full season.
- Seed rate and selection: use viable, certified seed of known varieties.

### **CLASS EVALUATION**

1. List the steps of land preparation.
2. Differentiate between direct sowing and transplanting.
3. Why is spacing important?
4. What determines the depth of sowing?

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: TEN

CLASS: SS1

SUBJECT: Agricultural Science

## **TOPIC: Farm Tools & Implements**

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### **Introduction**

- Farm tools are simple hand implements; implements are the larger, often animal- or machine-drawn equipment. This week covers the common field tools, their uses and maintenance.

### **Hand tools and uses**

- **Cutlass** — clearing, cutting; **hoe** — weeding, ridging; **spade** — digging; **mattock** — heavy digging and prying; **rake** — levelling, gathering; **garden fork** — loosening; **hand trowel** — transplanting; **watering can** — watering; **knapsack sprayer** — applying chemicals; **sickle/scythe** — harvesting; **wheelbarrow** — carrying loads.

### **Simple implements and machines**

- Animal-drawn implements: the plough (turns the soil), harrow (breaks clods) and ridger. Machine-drawn: the tractor with plough, harrow, planter, weeder and harvester — fast and efficient but costly and requires skill and spare parts.

### **Maintenance**

- Clean after use, oil metal parts, sharpen blades, replace worn handles, store under cover on a rack, and use each tool for its own purpose.

### **CLASS EVALUATION**

1. Name five hand tools and their uses.
2. What is the function of a plough?
3. State three rules of tool maintenance.
4. Give two advantages and two disadvantages of tractors on a small farm.

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: ELEVEN

CLASS: SS1

SUBJECT: Agricultural Science

## **TOPIC: Weed Control & Plant Protection**

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### **Introduction**

- Weeds are plants growing where they are not wanted. They compete with crops for water, nutrients, light and space, harbour pests and diseases, and reduce yields and quality.

### **Methods of weed control**

- **Cultural** — timely weeding, hoeing, crop rotation, close spacing (smothering), mulching, clean seed and clean tools.
- **Mechanical** — hand pulling, hoeing, ploughing, mowing.
- **Biological** — natural enemies of weeds (insects, fungi) — seldom used alone.
- **Chemical** — herbicides (selective, e.g. 2,4-D for broadleaf weeds; or non-selective), applied with care to avoid crop damage and pollution.

### **Crop pests and diseases**

- Pests: rodents, insects (stem borers, grasshoppers, aphids), birds and nematodes — controlled by chemicals (insecticides), resistant varieties, field sanitation and biological control. Diseases: fungi, bacteria and viruses — control by resistant varieties, crop rotation, clean seed and fungicides.

### **CLASS EVALUATION**

1. Define a weed.
2. State three effects of weeds on crops.
3. Differentiate between cultural and chemical weed control.
4. Name two crop pests and how they are controlled.

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: TWELVE

CLASS: SS1

SUBJECT: Agricultural Science

**TOPIC: Basic Farm Animal Production (Poultry & Small Ruminants)**

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**Introduction**

- Animal production supplies meat, eggs, milk and income. Poultry (chickens) and small ruminants (goats, sheep) are the easiest enterprises for a smallholder.

**Poultry management**

- Systems: intensive (battery cage, deep litter), semi-intensive and extensive (free range).
- Requirements: brooding (warmth, feed and water for chicks), clean litter, vaccination (Newcastle disease, Gumboro), good feed (mash or pellets), lighting and ventilation, and records.
- Breeds: broilers (meat), layers (eggs) — improved breeds (Isa Brown, Ross) outperform the local fowl in production.

**Goats and sheep**

- Breeds: West African Dwarf (trypanotolerant), Sokoto Red, Sahel goat; sheep — the Uda, Yankasa and West African Dwarf.
- Management: housing (raised, ventilated pens), feeding (grazing + supplements, mineral licks), watering, deworming and vaccination against peste des petits ruminants (PPR), and control of external parasites.

**CLASS EVALUATION**

1. State two systems of poultry keeping.
2. Why are chicks brooded?
3. Name two diseases of poultry and their control.
4. Why is the West African Dwarf goat popular in the south?